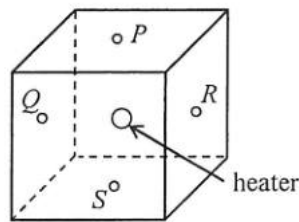


Book 1 section 1.3: DSE multiple choice

2020 Q1

1. A heater is installed at the centre of a fully filled cubic water tank. Temperature sensors P , Q , R and S are fixed at the respective centres of the top, left, right and bottom surfaces of the tank.

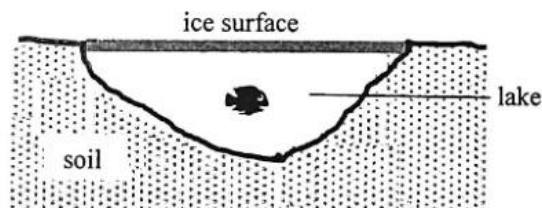


After the heater is switched on for a **short duration**, which pair of sensors below would indicate the largest temperature difference ?

- A. Q and R
- B. R and S
- C. Q and S
- D. P and R

2018 Q2

2. In some countries, the outdoor temperature can drop below $0\text{ }^{\circ}\text{C}$ in winter such that the lake surface becomes a thick layer of ice. However the water beneath the ice surface does not easily turn into ice and most aquatic life can survive during winter time.



Which statement below best explains this phenomenon ?

- A. The layer of ice provides good heat insulation.
- B. The freezing point of water below the ice surface is much lower than $0\text{ }^{\circ}\text{C}$.
- C. There is thermal energy transferred from the soil to the water in the lake.
- D. Ice releases latent heat when it melts.

2023 Q1

1. Which of the following statements involving heat transfer via radiation is/are correct ?
- (1) Objects hotter than the surroundings do not absorb heat by radiation.
 - (2) The silvery surfaces of survival blankets help the body absorb heat from the surroundings by radiation.
 - (3) The cooling fins of a car engine should preferably be black in colour.
- A. (1) only
 - B. (3) only
 - C. (1) and (2) only
 - D. (2) and (3) only

Book 1 section 1.3: DSE multiple choice

2026 Q1

1. Flasks X and Y are identical except that for their outer surfaces: X is black and Y is silvery. Each flask contains the same amount of water initially at $25\text{ }^{\circ}\text{C}$, sealed and equipped with a temperature sensor. The flasks are then placed in the same outdoor environment at $30\text{ }^{\circ}\text{C}$ with sunlight. After 20 minutes,
- A. the water in X is hotter as a black surface is a better absorber of radiation.
 - B. the water in X is hotter as a black surface is a better emitter of radiation.
 - C. the water in Y is hotter as a silvery surface is a poorer absorber of radiation.
 - D. the water in Y is hotter as a silvery surface is a poorer emitter of radiation.